Generalised Linear Mixed Models

for data of multiple species

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Questions so far?

Random-effects

When to include a random effect:

- Unobserved effect
- To account for pseudo replication
- Nuisance
- To induce correlation
- Shrinkage

Why "random" effects?

"Random" is in contrast to "fixed":

- Fixed effects are **maximised**: the likelihood is optimised directly over $\hat{\beta}$
- Random effects are **marginalised**: $\mathbf{u}$ is integrated out of the likelihood

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Random effects are stochastic: like the data, they come from a distribution $\mathbf{u} \sim \mathcal{N}(\mathbf{0}, \mathbf{\Sigma})$. A **mixed-effects model** contains both.

Our new likelihood

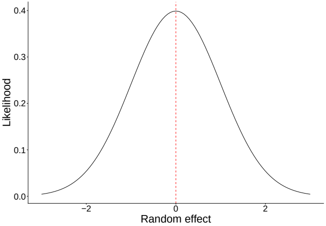
$$\mathcal{L}(\mathbf{y};\boldsymbol{\Sigma}) = \int \prod_{i}^{n} f(y_i|\mathbf{u})\mathcal{N}(\mathbf{u};\boldsymbol{\Sigma})d\mathbf{u} \tag{1}$$

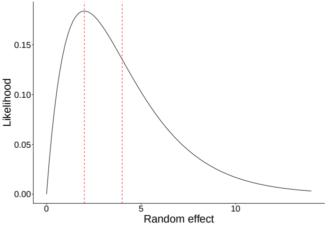
We approximate with:

- Penalized quasi-likelihood methods
- Adaptive GH quadrature
- Laplace approximation (gllvm)
- Variational approximations (gllvm)

Estimation and $\mathbf{u}|\mathbf{y}$

The posterior can be skewed for non-Gaussian responses (right); mode-based methods (Laplace) and mean-based methods (Variational) then give different results:





The mixed-effects model

$$g\{\mathrm{E}(\mathbf{y}|\mathbf{u})\} = \mathbf{X}\beta + \mathbf{Z}\mathbf{u} \tag{2}$$

1. Link-function
2. Conditional mean
3. Fixed effects design matrix
4. Random effects design matrix

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2. Conditional mean
3. Fixed effects parameter vector
4. Random effects parameter vector

There are many R-packages

- nlme
- lme4
- glmmTMB (or glmmADMB)
- sdmTMB
- MASS
- glmmML
- repeated
- glmm
- hglm
- spaMM
- gllvm
- mcmcGLMM
- INLA
- inlabru
- MCMC frameworks (JAGS, STAN, NIMBLE, greta)

`glmmVA` (gllvm)

- Long-format interface for multispecies data
- Species-specific fixed and random effects
- `lme4`-style formula for random effects
- Uses variational approximation (same as `gllvm`)
- Familiar syntax; bridges VGLMs to VGLMMs

Random effects in `gllvm`

In the `gllvm` `R`-package there are three formula interfaces:

- `glmmVA`
- `row.eff`
- `formula`
- `lv.formula`

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- `glmmVA` : for ordinary GLM(M)s
- `row.eff` : for species-common fixed/random effects
- `formula` : for species-specific fixed/random effects
- `lv.formula` : for effects in the ordination

Random effects syntax in `gllvm::glmmVA`

Now some examples of how it works in `R`. Generally:

`formula = ~ (0 + covariate | grouping)`

(the 0 is to omit an intercept term)

"Nested":

`formula = ~ (1|a/b)` is the same as `formula = ~ (1|a:b + b)`

"Crossed":

`y ~ (1|a) + (1|b)`

Effects within the same brackets are assumed to be correlated

Random effects syntax: implications

LHS determines the size of $\boldsymbol{\Sigma}$; RHS the number of draws per $1 \ldots n$ site (p species).

Formula	$\boldsymbol{\Sigma}$	REs per site	Total REs
`(1|Site)`	1×1	1	n
`(0+Species|Site)`	$p \times p$	p	$p \cdot n$
`(Species|Site)`	$(p+1) \times (p+1)$	$p+1$	$(p+1) \cdot n$

GLMMs for community ecology

- Hierarchical study designs are easily accommodated
- Quantify variation in species environmental responses
- Rare species borrow strength from the community: shrinkage to the mean
- Variance components summarize community-level structure: ICC and repeatability
- Covariance components can summarize niche attributes
- Correlation relaxes vital assumptions; species independence

Some examples follow

What VGLMMs add

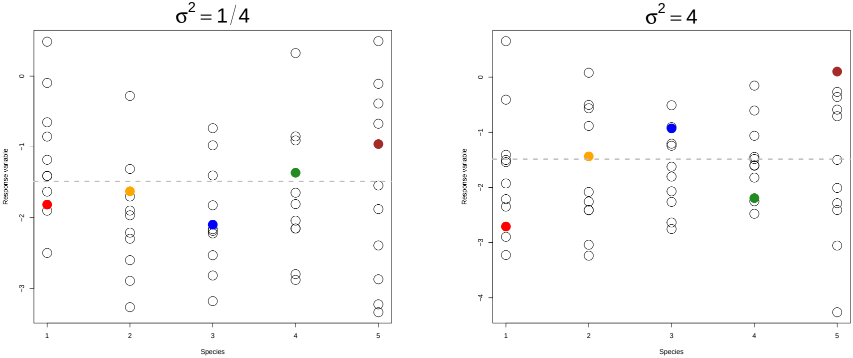
A VGLM estimates one fixed effect per species: it describes species' mean responses but cannot say how much they vary.

- **Variance components** (σ^2): how large is between-species variation in responses? Quantifies environmental filtering and niche variation.
- **Shrinkage**: rare species borrow strength from the community mean; estimates are regularized rather than fit freely to noise.
- **ICC and repeatability**: translate variance components into interpretable fractions. How much variation is explained by site? By species identity?

Variation in mean abundance

$$y_{ij} = \mu_\alpha + \alpha_j, \qquad \text{with } \alpha_j \sim \mathcal{N}(0, \sigma^2_\alpha)$$

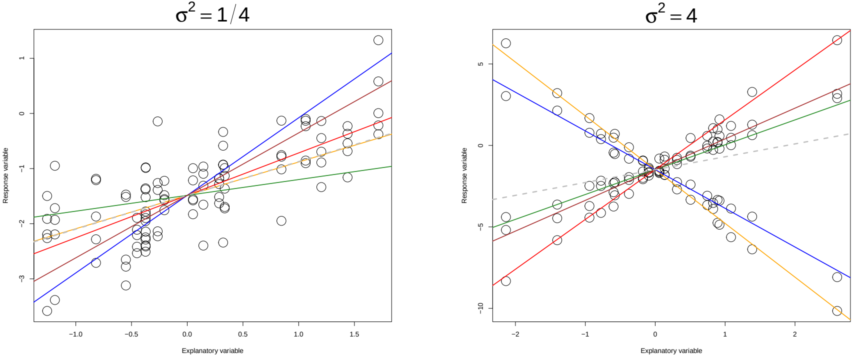
```
y ~ (1|Species)
```



Variation in environmental responses

$$y_{ij} = \mu_\alpha + x_i\mu_\beta + x_i\beta_j, \qquad \text{with } \beta_j \sim \mathcal{N}(0, \sigma^2_\beta)$$

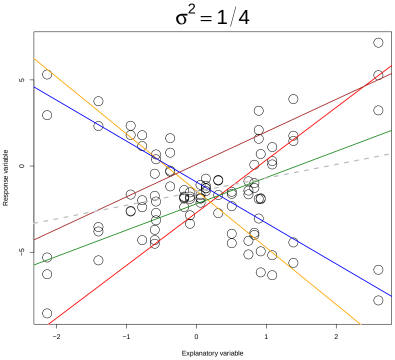
```
y ~ covariate + (0+covariate|Species)
```

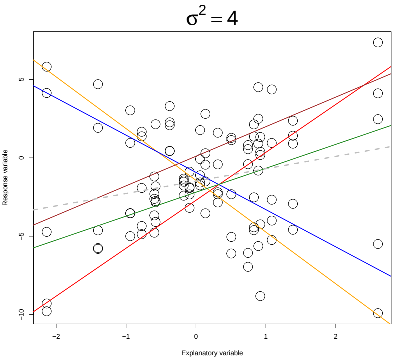


Variation of mean abundance and environmental responses

$$y_{ij} = \mu_\alpha + \alpha_j + x_i(\mu_\beta + \beta_j), \text{ with } \begin{pmatrix} \alpha_j \\ \beta_j \end{pmatrix} \sim \mathcal{N}\left(\begin{matrix} 0 \\ 0 \end{matrix}, \begin{matrix} \sigma^2_\alpha & 0 \\ 0 & \sigma^2_\beta \end{matrix} \right)$$

```
y ~ covariate + (covariate||Species)
```

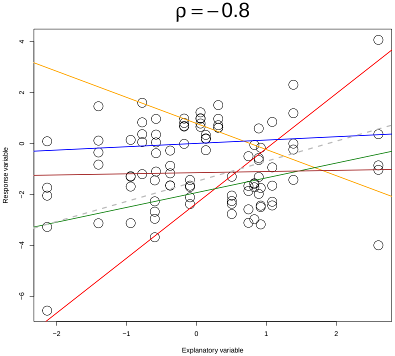


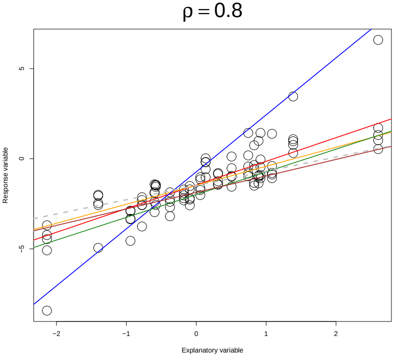


Correlation of mean abundance and environmental responses

$$y_{ij} = \mu_\alpha + \alpha_j + x_i(\mu_\beta + \beta_j), \begin{pmatrix} \alpha_j \\ \beta_j \end{pmatrix} \sim \mathcal{N}\left(\begin{matrix} 0 \\ 0 \end{matrix}, \begin{matrix} \sigma_\alpha^2 & \sigma_{\alpha,\beta} \\ \sigma_{\beta,\alpha} & \sigma_\beta^2 \end{matrix} \right)$$

```
y ~ covariate + (covariate|Species)
```





Number of levels

In typical mixed-effects models, the rule of thumb is 5 levels to estimate a variance

- Here, the species are our "levels" (so we usually have many)
- Few species: not a good correlation estimate
- Fewer species: not a good variance estimate

Species correlation: from σ^2 to Σ

$\beta_j \sim \mathcal{N}(\mu_\beta, \sigma^2)$ independently for each species j:

- All species share the same variance σ^2
- All species share the same mean μ_β
- Species responses are independent draws from this distribution

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- Species responses are independent draws from this distribution

To model **correlation** between species responses, we need a full covariance matrix:

$$\boldsymbol{\beta} \sim \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma}), \quad \boldsymbol{\Sigma} \text{ is } p \times p \tag{3}$$

- $\boldsymbol{\Sigma}$ allows cov$(\beta_j, \beta_k) \neq 0$: some species respond similarly
- Estimating $\boldsymbol{\Sigma}$ requires $p(p+1)/2$ parameters: infeasible when $n \ll p$

Example 1

Remember: we have 14 species. Enough for variances, not for correlations.

In the previous presentation, we saw that NO3 had no statistically significant effect. A small estimated variance produces strong **shrinkage** toward the community mean: species-specific estimates are regularized rather than fit freely to noisy data.

Wetland macroinvertebrates

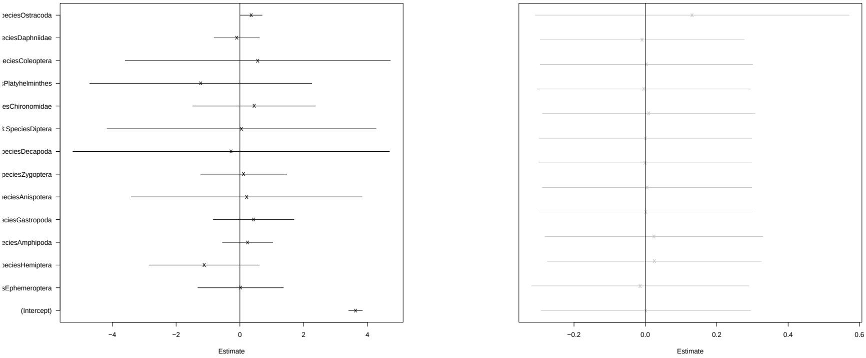
```
model1 <- glmmVA(Count~NO3:Species,
          family = "negative.binomial", data = long)
model2 <- glmmVA(Count~(0+NO3|Species),
          family = "negative.binomial", data = long)
```

Wetlands: summary

```
##
## Call:
## glmmVA(formula = Count ~ (0 + NO3 | Species), data = long, family = "negative.binomial")
##
## Family:  negative.binomial
##
## AIC:  2886.975 AICc:  2887.025 BIC:  2899.503 LL:  -1440.5 df:  3
##
##
## Formula:  ~(0 + NO3 | Species)
##
##  Random effects:
##  Name        Variance Std.Dev
##  NO3|Species 0.0232   0.1525
```

`Random effects` gives $\hat{\sigma}^2 = 0.023$ for the NO3 slope, near-homogeneous species responses.

Results: plots



Random slopes (right) are pulled toward zero relative to fixed effects (left): species with limited data borrow strength from the community mean rather than tracking noise.

glmmVA: random slope for all species

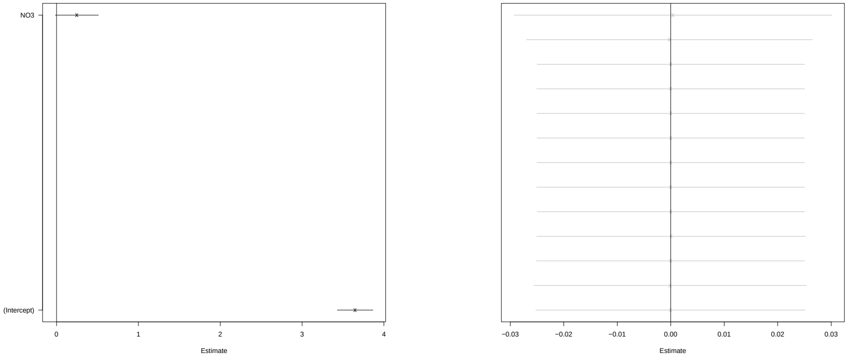
$$\beta_j \sim \mathcal{N}(\mu_\beta, \sigma^2_\beta), \quad j = 1, \dots, p$$

```
model2b <- glmmVA(Count ~ NO3 + (0+NO3|Species),
                  family = "negative.binomial", data = long)
```

```
## NO3|Species
##  0.01276583
```

- Each species gets its own NO3 effect
- All effects drawn from the same $\mathcal{N}(\mu_\beta, \sigma^2_\beta)$
- Small $\hat{\sigma}^2$: species NO3 responses are near-homogeneous; consistent with the non-significant fixed effect
- Same model as `formula = ~(0+NO3|1)` in `gllvm`

Average community NO3 response



Community-mean NO3 effect (left); species-specific deviations from that mean (right). All deviations are near zero: species respond similarly to NO3, consistent with the small estimated $\hat{\sigma}^2$.

Wetlands: two effects

Let's add more effects

```
model3 <- glmmVA(Count ~(1|Species)+(0+NO3|Species)+(0+SO4|Species),
                 family = "negative.binomial", data = long)
```

Variance components

What fraction of between-species variation is due to SO4?

```
(sigma <- model3$params$sigma)
```

```
## (Intercept)|Species          NO3|Species          SO4|Species
##        1.5524420051         0.0006437364         0.4223646539
```

```
sigma[grep("SO4", names(sigma))] / sum(sigma)
```

```
## SO4|Species
##   0.2138068
```

SO4 is a stronger environmental filter than NO3: sulphate availability structures macroinvertebrate community composition.

Full species covariance: not yet

```
# 13 species on LHS: need a 13x13 covariance from 14 sites
m_full <- glmmVA(Count ~ NO3 + (0+Species|Site),
                 data = long, family = "negative.binomial",
                 sd.errors = FALSE)
```

$13 \times 13 \rightarrow 13 * (13+1)/2 = 91$ variance parameters from $14 * 13 = 182$ observations: computationally expensive.

Often doesn't converge, or is terribly slow.

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Ordination reduces residual species covariance to a few latent dimensions. More tomorrow.

From glmmVA to gllvm

- `glmmVA` calls `gllvm` internally wrapping `row.eff`
- The `formula` interface provides species-specific random effects in wide format (with mean effects)
- VGLMMs: the covariance structure is shared across all species, but own dispersion
- Potentially species-specific response types

glmmVA is a wrapper around row.eff

glmmVA internally calls `gllvm(..., row.eff = formula, ...)`:

```
glmmVA(Count ~ Species + (0+NO3|Species), ...)
# strips the response, passes the formula as row.eff:
gllvm(y, studyDesign=X, row.eff=~(0+NO3|Species), ...)
```

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gllvm(y, studyDesign=X, row.eff=~(0+NO3|Species), ...)
```

gllvm's `formula` interface is the wide-format equivalent:

```
gllvm(y, X=X, formula=~(0+NO3|1), beta0com=TRUE, ...)
```

`|Species` (long format) $\equiv$ `|1` (wide format): one $\mathcal{N}(\mu, \sigma^2)$ for all species.
`formula` gives species-specific β_j; `row.eff` gives site-level effects shared across species.

gllvm: the VGLMM engine

glmmVA covers the long-format case. gllvm is the underlying engine and exposes the full model directly.

- Wide-format (matrix) data: species $\times$ sites
- Species-specific distributions and dispersion parameters
- All three formula interfaces simultaneously

```
# Long format: single data frame
glmmVA(Count ~ Species + (0+NO3|Species), data = long, ...)
# Wide format: response matrix and covariate matrix, no data= argument
gllvm(y, X = X, formula = ~(0+NO3|1), ...)
```

For VGLMMs we use gllvm directly from here on

`row.eff`: common effects in `gllvm`

In `glmmVA`, the formula interface covers all random effects. In `gllvm`, `row.eff` is the interface to effects for the rows of the response matrix.

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- `row.eff` is a mixed-effects formula
- `row.eff = ~1` omits the common effects
- `row.eff = "random"` incorporates row-specific random effects
- `row.eff = (1|group) + NO3` is a random effect and a fixed effect
- Can also incorporate spatial or temporal random effects

Example 2

For the Swiss birds, we have 56 species, so we can do a little more.

In the previous presentation, we saw that the effect of slope was statistically significant. Treated as random, it is not.

Example 2: fitting correlated effects

```
model6 <- gllvm(y, X = X, formula = ~(bio_1+bio_12|1),
                family = binomial(), num.lv = 0,
                beta0com = TRUE)
```

Birds that prefer warm conditions also prefer drier conditions?

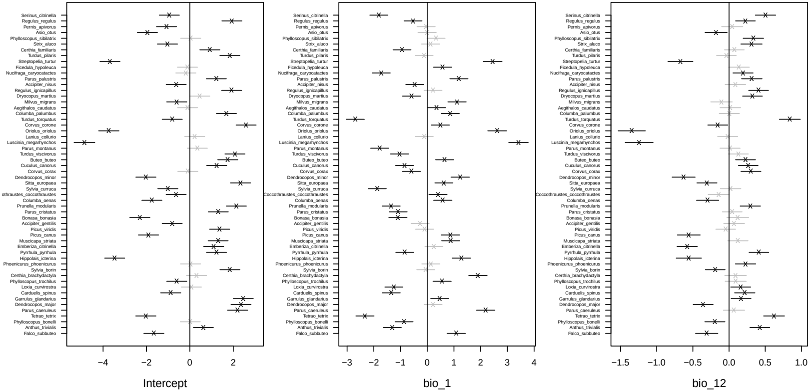
Example 2: summary

```
...
## Formula:  ~(bio_1 + bio_12 | 1)
## LV formula:  ~ 0
## Row effect:  ~ 1
##
##  Multispecies random effects:
##  Name      Variance Std.Dev Corr
##  Intercept 3.0005   1.7322
##  bio_1     1.5280   1.2361  -0.2512
##  bio_12    0.1570   0.3963   0.4618 -0.6865
##
## Coefficients predictors:
##           Estimate Std. Error z value Pr(>|z|)
## Intercept -1.35202    0.23198  -5.828  5.6e-09 ***
## bio_1      0.41878    0.16586   2.525  0.01157 *
## bio_12    -0.19071    0.05407  -3.527  0.00042 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
...
```

Indeed! Coupled climatic filtering in community assembly.

Example 2: coefficient plot

```r
randomCoefplot(model6)
```



Sorting along temperature and precipitation: joint filtering.

ICC

- How much does community composition vary between sites relative to within sites?
- How consistent is a species' environmental response across contexts?
- Niche concepts, ICC and repeatability answer these questions from variance components directly.

ICC and repeatability in community ecology

Common in behavioral ecology and animal breeding.

- VGLMMs produce variance components naturally: ICC and repeatability give these an ecological interpretation
- Neither requires additional modelling: they follow directly from $\hat{\boldsymbol{\Sigma}}$ and $\hat{\sigma}^2$
- Applicable wherever there is a grouping structure: sites, reserves, years

BIOLOGICAL REVIEWS Cambridge Philosophical Society

Repeatability for Gaussian and non-Gaussian data: a practical guide for biologists

Shinichi Nakagawa, Holger Schielzeth

First published: 21 June 2010 | https://doi.org/10.1111/j.1469-185X.2010.00141.x |

Intraclass correlation in community ecology

ICC: correlation between two observations from the same class:

$$\mathrm{ICC}_p = \frac{\sigma_p^2}{\sigma_1^2 + \sigma_p^2 + \cdots} = \mathrm{corr}(u_{jp}, u_{kp})$$

- **(1|Site)**: $\mathrm{ICC}_{\mathrm{site}}$: what fraction of variation is explained by site? High ICC: site effects dominate community composition; species co-respond to shared environmental conditions.
- **(1|Species)**: $\mathrm{ICC}_{\mathrm{sp}}$: what fraction is explained by species identity? High ICC: species differ strongly in mean abundance.
- Correlated random effects: total variance is sum$(\boldsymbol{\Sigma})$, not the diagonal alone; ICC is still computable but the off-diagonal must be included

Repeatability in community ecology

Repeatability is a special case of ICC where the denominator also includes the observation-level (residual) variance, the within-group variation attributable to sampling noise. For a binomial logit model (Nakagawa & Schielzeth 2010):

$$R = \frac{\sigma^2_{\text{group}}}{\sigma^2_{\text{group}} + \pi^2/3}$$

- **Site repeatability**: how similar is community composition on two plots from the same site? Answers: how strongly does site structure the community?
- **Species repeatability**: how consistent is a species' environmental response across contexts? Answers: how *fixed* is a species' niche across sites?

Example 3

59 wood-decaying fungal species on 1666 dead logs across 53 Scandinavian nature reserves (Abrego 2021). Dead wood volume, tree type, and reserve connectivity are key habitat drivers.

Logs (plots) nested within reserves (sites): a natural hierarchical structure for ICC and repeatability.

Wood-decaying fungi (Abrego 2021)

Received: 1 November 2021 | Accepted: 20 December 2021
DOI: 10.1111/1365-2745.13839
Journal of Ecology BRITISH ECOLOGICAL SOCIETY

RESEARCH ARTICLE

Traits and phylogenies modulate the environmental responses of wood-inhabiting fungal communities across spatial scales

Nerea Abrego[1,2] | Claus Bässler[3,4] | Morten Christensen[5] | Jacob Heilmann-Clausen[6]

Does community composition differ more *within* reserves (log-level variation) or *between* reserves (reserve-level variation)?

Fungi: model

```
m_rep <- gllvm(fy, X=fX,
               studyDesign=data.frame(RESERVE=fX$RESERVE),
               formula=~(0+DBH.CM|1),
               row.eff=~(1|RESERVE),
               family="binomial", num.lv=0,
               beta0com=FALSE)
```

- `formula`: species-specific random DBH slope (σ^2_{DBH}); species vary in how strongly they filter by log size
- `row.eff`: reserve-level random effect (σ^2_{res}); shared reserve context affects all species

Species repeatability

How consistent is a species' DBH (log diameter) response across reserves?

```
sigma_dbh <- as.numeric(m_rep$params$sigmaB)
sigma_dbh / (sigma_dbh + pi^2/3)
```

```
## [1] 0.01958321
```

A high value would mean a species' sensitivity to log diameter is a stable niche attribute; a low value means it varies unpredictably across reserves.

Site repeatability

How similar is community composition on two logs from the same reserve?

```r
sigma2_site <- m_rep$params$sigma["(Intercept)|RESERVE"]^2
sigma2_site / (sigma2_site + pi^2/3)
```

```
## (Intercept)|RESERVE
##          0.01487257
```

A high value means reserve identity is the dominant driver of community composition; a low value means within-reserve log-to-log variation dominates. Both use $\pi^2/3$ as the latent-scale residual for binomial data.

End